

2 (a) the number of atoms in 12 g of carbon-12	M1	
	A1	[2]
(b) (i) amount = $3.2/40$ = 0.080 mol	A1	[1]
(ii) $pV = nRT$ $p \times 210 \times 10^{-6} = 0.080 \times 8.31 \times 310$ $p = 9.8 \times 10^5 \text{ Pa}$ <i>(do not credit if T in °C not K)</i>	C1	
	A1	[2]
(iii) either $pV = 1/3 \times Nm \langle c^2 \rangle$ $N = 0.080 \times 6.02 \times 10^{23} (= 4.82 \times 10^{22})$ <u>and</u> $m = 40 \times 1.66 \times 10^{-27} (= 6.64 \times 10^{-26})$ $9.8 \times 10^5 \times 210 \times 10^{-6} = 1/3 \times 4.82 \times 10^{22} \times 6.64 \times 10^{-26} \times \langle c^2 \rangle$ $\langle c^2 \rangle = 1.93 \times 10^5$ $c_{\text{RMS}} = 440 \text{ m s}^{-1}$	C1	
	C1	
	A1	[3]
or $Nm = 3.2 \times 10^{-3}$ $9.8 \times 10^5 \times 210 \times 10^{-6} = 1/3 \times 3.2 \times 10^{-3} \times \langle c^2 \rangle$ $\langle c^2 \rangle = 1.93 \times 10^5$ $c_{\text{RMS}} = 440 \text{ m s}^{-1}$	(C1)	
	(C1)	
	(A1)	
or $1/2 m \langle c^2 \rangle = 3/2 kT$ $1/2 \times 40 \times 1.66 \times 10^{-27} \langle c^2 \rangle = 3/2 \times 1.38 \times 10^{-23} \times 310$ $\langle c^2 \rangle = 1.93 \times 10^5$ $c_{\text{RMS}} = 440 \text{ m s}^{-1}$	(C1)	
	(C1)	
	(A1)	
<i>(if T in °C not K award max 1/3, unless already penalised in (b)(ii))</i>		

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